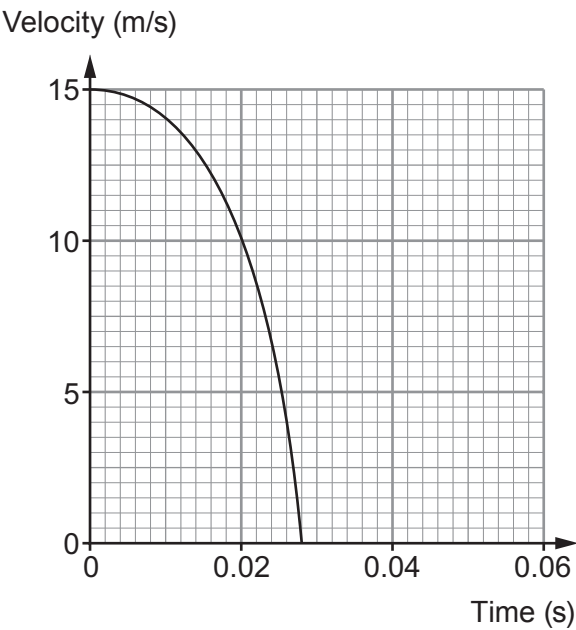


2. The safety of car drivers and passengers in the event of a collision has been the subject of an enormous amount of research in the last 50 years. As a result, the design of cars has developed to improve the safety of the occupants in the event of a head-on collision.

A crash dummy is positioned in the driving seat of a car, which is directed, under test conditions, into a solid concrete wall. The graph below shows the velocity of the dummy from the moment the car makes contact with the wall.



- (a) State Newton’s third law of motion and explain how it applies to the car and the wall in this collision. [3]

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Examiner
only

- (b) (i) Use the graph opposite and equations from page 2 to calculate the mean resultant force on the dummy of mass 85 kg during the collision.
Give your answer to two significant figures. [3]

Mean resultant force = N

- (ii) Use the graph opposite to explain whether the resultant force is constant during the collision. [2]

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- (c) Explain, **in terms of the work done**, how a crumple zone at the front of the car would improve the safety of the driver in a head-on collision. [2]

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